Experiment No. 4

Experiment Title	To implement Classification algorithms (Decision tree and Naïve Bayes' algorithms) using Python
Course Outcomes	CO4: Implement the appropriate data mining methods like classification, clustering or Frequent Pattern mining on large data sets. CO5: Define and apply metrics to measure the performance of various data mining algorithms.
Date of Preparation	
Date of Submission	
Signature & Grade	

1. Aim

To implement the Decision Tree and Naïve Bayes classification algorithms in Python (scikit-learn) on an airline passenger dataset, and to compare them using accuracy, precision, recall, F1-score, the confusion matrix, the ROC curve with AUC, and cross-validation.

2. Theory

2.1 Classification

Classification is a supervised data mining technique. A model is trained on records whose class is known (the training set) and is then used to predict the class of new records. Here the class is 'satisfaction' and a passenger is either 'satisfied' or 'neutral or dissatisfied'.

2.2 Decision Tree

A decision tree repeatedly splits the data on the attribute and value that best separates the classes. With the entropy criterion used here, the best split is the one with the highest information gain. Each internal node is a test, each branch an outcome and each leaf a class. Trees are easy to read, but a tree that is too deep memorises the training data (overfitting), so the maximum depth has to be chosen carefully.

2.3 Naïve Bayes

Naïve Bayes applies Bayes' theorem, $P(C|X) = P(X|C) \times P(C) / P(X)$, and assumes that all attributes are independent of each other given the class. Gaussian Naïve Bayes assumes that each numeric attribute follows a normal (Gaussian) distribution within each class. It is very fast, but the independence assumption is rarely perfectly true.

2.4 Evaluation Measures

Measure	Meaning
Accuracy	Share of test records classified correctly.
Precision	Of the records predicted 'satisfied', the share that really are satisfied.
Recall	Of the records that really are satisfied, the share that were found.
F1-score	Harmonic mean of precision and recall.
Confusion matrix	Table of actual classes against predicted classes.
ROC curve and AUC	True positive rate against false positive rate; AUC near 1 means good separation of the classes.
Cross-validation	The training data is split into folds; the model is trained on some folds and tested on the rest, to check that the result is stable.

3. Dataset and Tools

Dataset: Airline Passenger Satisfaction (Kaggle). The file train.csv (103,904 records) is used to train the models and to choose the tree depth. The separate file test.csv (25,976 records) is kept aside and used only for the final evaluation. There are 22 feature columns (passenger details, flight details and 14 service ratings from 0 to 5) and the target column 'satisfaction'.

Source: kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

Tools: Python 3 in Google Colab with pandas, NumPy, Matplotlib, Seaborn and scikit-learn.

4. Implementation

4.1 Prepare the Data

The target was converted to 0 and 1, and the row index and id columns were dropped. All preprocessing was placed inside a pipeline so that it is learned from the training data only: missing numeric values are filled with the median, missing text values with the most frequent value, and text columns are one-hot encoded. Five-fold stratified cross-validation was used inside the training data.

```
cat_steps = Pipeline([('imputer', SimpleImputer(strategy='most_frequent')),
                      ('encoder', OneHotEncoder(handle_unknown='ignore'))])
preprocessor = ColumnTransformer([
    ('numeric', SimpleImputer(strategy='median'), num_cols),
    ('categorical', cat_steps, cat_cols)])
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
```

```
Training records: 103904 | Test records: 25976
Original feature columns: 22
Missing feature values: train = 310 , test = 83
Training class balance (%):
satisfaction
neutral or dissatisfied    56.7
satisfied                  43.3
Preprocessing will be fitted inside each training fold.
```

Figure 1: Size of the training and test data, missing values and class balance

There are 103,904 training records and 25,976 test records with 22 feature columns. The training data has 310 missing values and the test data has 83. The training data is 56.7% neutral or dissatisfied and 43.3% satisfied.

4.2 Choose the Tree Depth

```
for depth in range(1, 16):
    tree = DecisionTreeClassifier(criterion='entropy', max_depth=depth,
                                  random_state=42)
    pipe = Pipeline([('preprocess', clone(preprocessor)), ('model', tree)])
    scores = cross_val_score(pipe, X_train, y_train, cv=cv, scoring='accuracy')
```

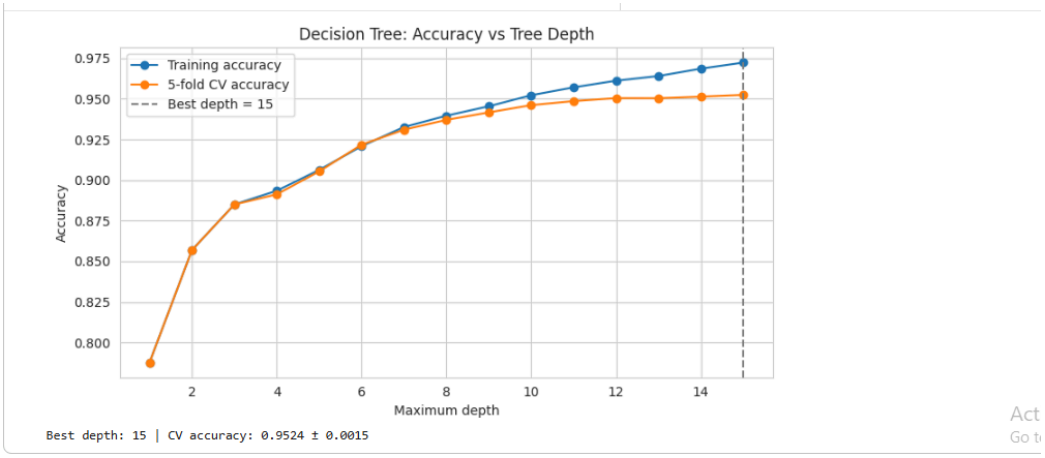


Figure 2: Training and cross-validation accuracy for tree depths 1 to 15

Training accuracy keeps rising with depth, while cross-validation accuracy levels off at about 0.95 after depth 12. The curves separate after depth 6, which shows that deeper trees start to memorise the training data. The best depth was 15 (cross-validation accuracy 0.9524 ± 0.0015), the largest depth tested, although the gain over depth 12 is very small.

4.3 Train the Decision Tree

```
tree = DecisionTreeClassifier(criterion='entropy', max_depth=best_depth,
                              random_state=42)
dt = Pipeline([('preprocess', clone(preprocessor)), ('model', tree)])
dt.fit(X_train, y_train)
```

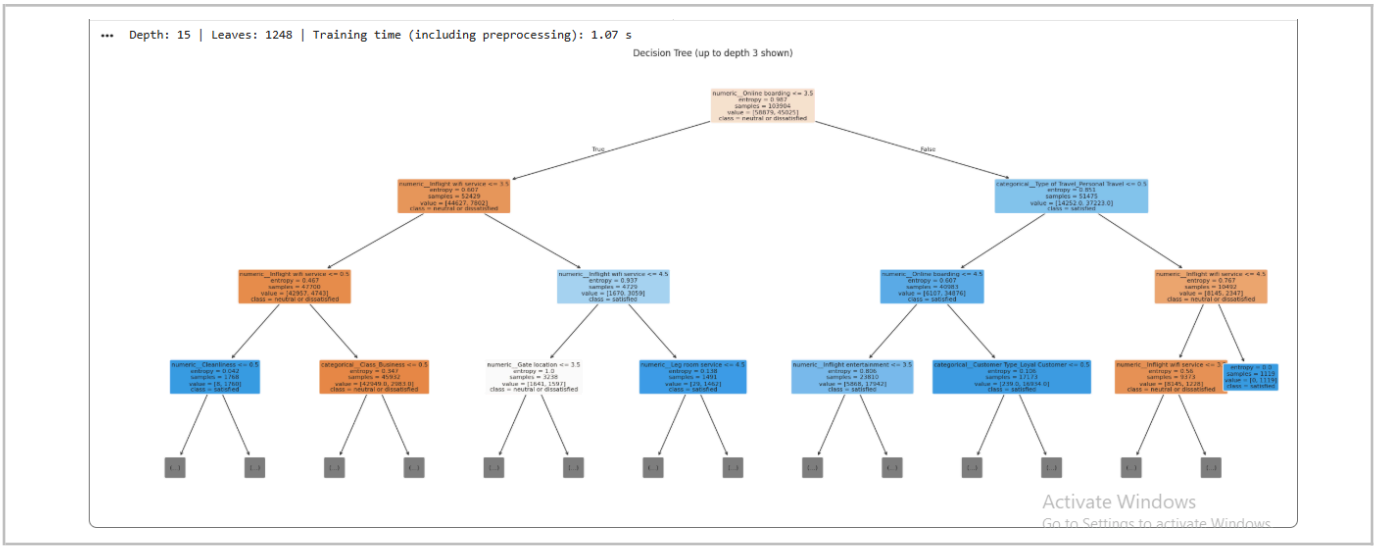


Figure 3: The trained Decision Tree (first 3 levels shown)

The final tree has depth 15 and 1,248 leaves and trained in 1.07 seconds (including preprocessing). The root test is Online boarding ≤ 3.5. Passengers with low ratings (52,429 of them) are mostly neutral or dissatisfied (44,627 against 7,802), and Inflight wifi service is the next test. Passengers with Online boarding above 3.5 (51,475) are mostly satisfied, and the next test is Type of Travel: business travellers in this group are mostly satisfied (34,876 of 40,983), while personal travellers are mostly not (8,145 of 10,492).

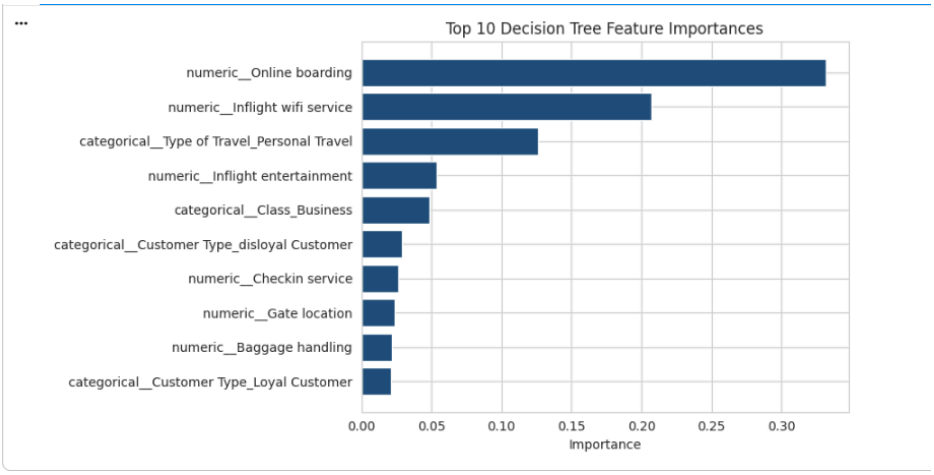


Figure 4: The ten most important features of the Decision Tree

Online boarding (about 0.33), Inflight wifi service (about 0.21) and Personal Travel (about 0.13) are the three most important features, followed by Inflight entertainment and Business class.

4.4 Train the Naïve Bayes Classifier

```
nb = Pipeline([('preprocess', clone(preprocessor)), ('model', GaussianNB())])
nb.fit(X_train, y_train)
```

```
Naïve Bayes training time (including preprocessing): 0.45 s
neutral or dissatisfied: 0.567
satisfied: 0.433
```

Figure 5: Naïve Bayes training time and class prior probabilities

Naïve Bayes trained in 0.45 seconds. Its prior probabilities, learned from the training data, are 0.567 for neutral or dissatisfied and 0.433 for satisfied.

4.5 Evaluate on the Test Set

```
dt_pred, nb_pred = dt.predict(X_test), nb.predict(X_test)
print(classification_report(y_test, dt_pred, target_names=CLASS_NAMES, digits=3))
confusion_matrix(y_test, dt_pred)
```

```
DECISION TREE
                       precision    recall  f1-score   support

neutral or dissatisfied     0.951     0.967     0.959     14573
            satisfied       0.957     0.936     0.946     11403

             accuracy                          0.953     25976
            macro avg       0.954     0.952     0.953     25976
         weighted avg       0.954     0.953     0.953     25976

NAÏVE BAYES
                       precision    recall  f1-score   support

neutral or dissatisfied     0.856     0.869     0.862     14573
            satisfied       0.829     0.813     0.821     11403

             accuracy                          0.844     25976
            macro avg       0.842     0.841     0.842     25976
         weighted avg       0.844     0.844     0.844     25976
```

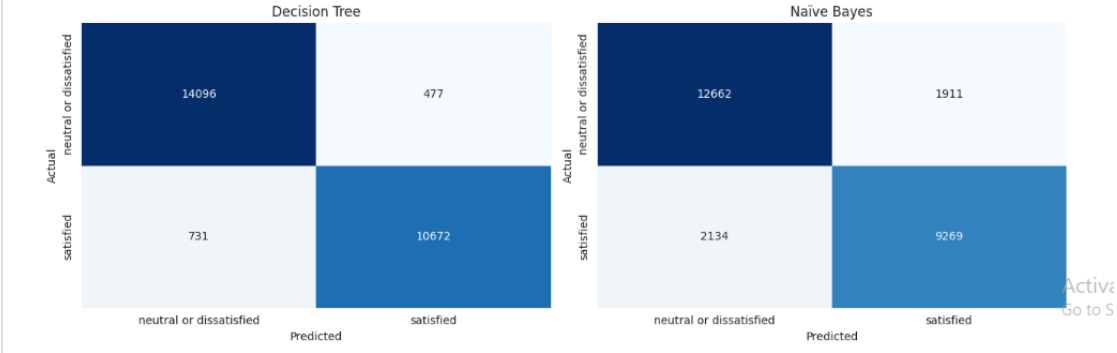


Figure 6: Classification reports and confusion matrices on test.csv

The Decision Tree classified 95.3% of the 25,976 test passengers correctly, with a precision of 0.957 and a recall of 0.936 for satisfied passengers. It made 1,208 errors (4.65%): 477 neutral or dissatisfied passengers were predicted as satisfied and 731 satisfied passengers were predicted as neutral or dissatisfied. Naïve Bayes classified 84.4% correctly, with a precision of 0.829 and a recall of 0.813 for satisfied passengers, and made 4,045 errors (15.57%): 1,911 and 2,134 of the two kinds. Both models find the neutral or dissatisfied class slightly better than the satisfied class.

4.6 ROC Curves

```
dt_prob = dt.predict_proba(X_test)[:, 1]
fpr, tpr, _ = roc_curve(y_test, dt_prob)
roc_auc_score(y_test, dt_prob)
```

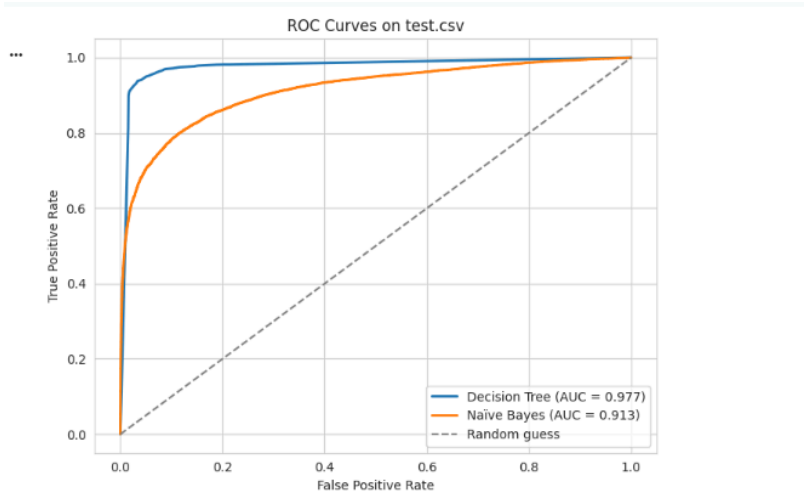


Figure 7: ROC curves of the two classifiers on test.csv

The Decision Tree curve rises steeply and stays close to the top-left corner (AUC 0.977). The Naïve Bayes curve is lower (AUC 0.913) but is still well above the diagonal line that represents random guessing.

4.7 Compare the Classifiers

Test metrics: positive class = satisfied

	Accuracy	Precision	Recall	F1-score	AUC	5-fold CV accuracy (mean)	5-fold CV accuracy (std)	Training time (s)
Decision Tree	0.9535	0.9572	0.9359	0.9464	0.9768	0.9524	0.0015	1.0296
Naïve Bayes	0.8443	0.8291	0.8129	0.8209	0.9135	0.8489	0.0032	0.4488

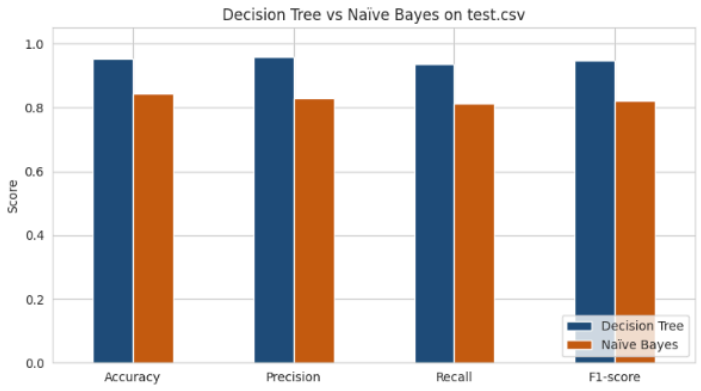


Figure 8: Comparison table and bar chart of the two classifiers

The Decision Tree is better on every test metric. Its cross-validation accuracy (0.9524 ± 0.0015) agrees closely with its test accuracy (0.9535), and the same holds for Naïve Bayes (0.8489 ± 0.0032 and 0.8443), so the results are stable. The cross-validation score of the tree was also used to choose its depth, so the independent test metrics are used for the final comparison.

5. Result Comparison

Measure (test.csv)	Decision Tree	Naïve Bayes
Accuracy	0.9535	0.8443
Precision (satisfied)	0.9572	0.8291
Recall (satisfied)	0.9359	0.8129
F1-score (satisfied)	0.9464	0.8209
AUC	0.9768	0.9135
Misclassified records	1,208 (4.65%)	4,045 (15.57%)
5-fold CV accuracy (mean ± std)	0.9524 ± 0.0015	0.8489 ± 0.0032
Training time (including preprocessing)	1.03 s	0.45 s

6. Conclusion

The Decision Tree and Gaussian Naïve Bayes classifiers were implemented in Python on the Airline Passenger Satisfaction dataset. Trained on train.csv and evaluated on the separate test.csv, the Decision Tree reached an accuracy of 95.35% and an AUC of 0.977, while Naïve Bayes reached 84.43% and an AUC of 0.913. The Decision Tree was better in precision, recall and F1-score, and the cross-validation results agreed with the test results.

Naïve Bayes trained faster (0.45 s against 1.03 s) but was about 11 percentage points less accurate. A likely reason is its independence assumption, because many of the service ratings are related to each other. The Decision Tree also gives readable rules and feature importances, which showed that Online boarding, Inflight wifi service and Type of Travel matter most. This agrees with Experiment 3, where Online boarding had the strongest correlation with satisfaction. For this dataset, the Decision Tree is therefore the better classifier.